

NOT ON YOUR FORMULA SHEET — THE WHOLE LIST

Index: $1/x^n = x^{-n}$ · $\int x = x^{0.5}$ · $\sqrt[3]{x} = x^{1/3}$ · $x^0 = 1$

Logs: change of base $\log_b x = \lg x / \lg b$ · the swap $y = b^x \leftrightarrow x = \log_b y$

Lines: $y - y_1 = m(x - x_1)$ · parallel $m_1 = m_2$ · perpendicular $m_1 m_2 = -1$ · collinear = same gradient

Quadratics: discriminant $b^2 - 4ac$: > 0 two roots, $= 0$ one, < 0 none · $a > 0$ MIN, $a < 0$ MAX

Factor theorem: $f(a) = 0$ means $(x - a)$ is a factor

Triangles: SOH CAH TOA · **sine rule** $\sin A/a = \sin B/b$ · **cosine rule** $a^2 = b^2 + c^2 - 2bc \cos A$ · Pythagoras

Vectors: $|v| = \sqrt{(x^2 + y^2)}$ · $AB = B - A$ · quadrant placing

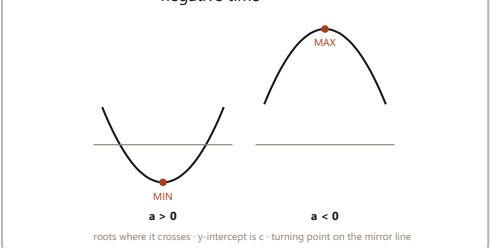
Matrices: order rows $\times$ cols · inside numbers must match

Calculus: $f_a/(bx+d) = (a/b)\ln(bx+d) + c \cdot d/x$ · $\ln f = f'/f$ · chain rule · $A \sin(Bx+C) \rightarrow AB \cos(Bx+C)$ · 2nd derivative: **+ve MIN**, **-ve MAX** · area below axis: split, drop sign, add

FOD: the three solutions $Ae^{-kt} / Ae^{kt} / c + Ae^{-kt}$

STUCK? WHAT IS IT ASKING FOR		
THE QUESTION SAYS	IT WANTS	GO TO
"dy/dx", "rate of change", "gradient function"	differentiate	DIFFERENTIATE
"maximum", "minimum", "turning point"	differentiate, =0, sub into ORIGINAL	TURNING POINTS
"area under", numbers on $\int$	integrate, top take bottom, no +c	INTEGRATE
"roots", "crosses the x-axis"	set $y = 0$	QUADRATICS
"y-intercept"	it IS c	free mark
"in terms of x"	that letter alone on the left	REARRANGING
"show your working"	every line earns; a bare wrong answer earns nothing	always write line 1

QUADRATICS		
ASKED FOR	MOVE	WORKED
y-intercept	it is c	$6x^2 + 14x - 12 \rightarrow -12$
roots, factorises	two numbers $\times$ to c, + to b	$(3x-2)(x+3) \rightarrow 2/3, -3$
roots, does not	name a,b,c with signs, sheet formula	$196 + 288 = 484, \sqrt{} = 22$
max or min	sign of a only. +ve MIN, -ve MAX	$a = -6 \rightarrow$ maximum
how many roots	$b^2 - 4ac$: > 0 two, $= 0$ one, < 0 none	$16 - 32 = -16 \rightarrow$ none
line of symmetry	$x = -b/2a$. write it as x = ...	roots 3, 5 $\rightarrow x = 4$
turning point	that x, then sub into the ORIGINAL	$\rightarrow (4, -1)$
$(x+q)^2 + r$	$q = b/2, r = c - q^2$	$x^2 - 10x + 3 \rightarrow (x-5)^2 - 22$
$p(x+q)^2 + r, a \neq 1$	q = b/2a, r = c - b^2/4a. do NOT halve b first	$2x^2 + 8x + 15 \rightarrow 2(x+2)^2 + 7$
given as () ()	EXPAND before reading a, b, c	$(3x-2)(4-2x) \rightarrow a = -6$
y not alone	rearrange to $y = \dots$ first	$y + 19x - x^2 = 10x - 10 \rightarrow x^2 - 9x - 10$
"height after 2 s"	first substitute	$6 + 260 - 20 = 246$ m
"max height"	$x = -b/2a$, then sub back	$x = 13 \rightarrow 851$ m
"hits the ground"	set $h = 0$, discard negative time	$\rightarrow 2.21$ s



STRAIGHT LINES AND COORDINATE GEOMETRY		
GIVEN / ASKED	MOVE	WORKED
two points $\rightarrow$ gradient	up over along, SAME order top and bottom	$(2, 0), (4, 10) \rightarrow 5$

equation not tidy	divide EVERY term into $y = mx + c$ first	$3y = 6x + 24 \rightarrow y = 2x + 8$
x-intercept	set $y = 0$	$0 = 2x + 8 \rightarrow -4$
gradient + a point	$y - y_1 = m(x - x_1)$	$m = 4, (6, -2) \rightarrow y = 4x - 26$
two points, no gradient	gradient first, then as above	$(2, 4), (5, -2) \rightarrow y = -2x + 8$
parallel?	same m. rearrange each one first	$6x - 2y = 10 \rightarrow m = 3 \checkmark$
perpendicular / normal	$m_2 = -1/m_1$ flip and change sign	$m = 4/5 \rightarrow -5/4$
collinear?	gradient PQ = gradient QR	all $-1/2 \rightarrow$ yes
perpendicular bisector	midpoint AND $-1/m$. both halves	$A(4, 3) B(-2, 5) \rightarrow y = 3x + 1$
right-angled triangle?	two gradients multiply to -1	$(4/3)(-3/4) = -1 \checkmark$
area of a triangle	$\frac{1}{2} x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)$	$\rightarrow 10$
sketch a region	dotted for $<$, $>$, solid for $\leq$, $\geq$. test (0,0)	true $\rightarrow$ shade that side

LOGS AND EXPONENTIALS		
LOOKS LIKE	MOVE	WORKED
$\log_b 8 = 3$	read aloud: what power of b gives 8	$b^3 = 8 \rightarrow b = 2$
$\log_a b = 2$	the answer IS the power	$4^2 \rightarrow b = 16$
$\log_2(\log_3 x) = 1$	peel the OUTSIDE first	$\log_3 x = 2 \rightarrow x = 9$
$\lg x = \lg 3 + \lg 8$	added $\rightarrow$ insides multiply	$x = 24$
$\lg x = \lg 40 - \lg 5$	subtracted $\rightarrow$ insides divide	$x = 8$
$\log_2 1 + \log_5 5^4$	$\log 1 = 0$; power comes to the front	$0 + 4 = 4$
$\log_2 10$, no button	change of base: $\lg 10 / \lg 2$	$= 3.32$
$5^x = 8$	$\lg$ both sides, power down, divide	$\lg 8 / \lg 5 = 1.29$
$3^{x+1} = 12$	the WHOLE bracket comes down, then -1	$2.26 - 1 = 1.26$
3, 12, 48, 192, ...	divide consecutive terms for the ratio	$\times 4 \rightarrow 768$
$\times 3$ every 20 s, after 1 min	count the STEPS $60/20 = 3$	$12 \times 3^3 = 324$

PARTIAL FRACTIONS AND FACTOR THEOREM		
BOTTOM LOOKS LIKE	SET UP AS	ANSWER
$(x-1)(x+8)$ different	$A/(x-1) + B/(x+8)$	kill one bracket at a time
$(x+5)^2$ squared alone	$A/(x+5) + B/(x+5)^2$	$10/(x+4) - 5/(x+4)^2$
$(x+6)(x-1)^2$	$A/(x+6) + B/(x-1) + C/(x-1)^2$	$8/(x+6) - 1/(x-1) + 4/(x-1)^2$
$x(x^2 + x + 7)$ will not factorise	$A/x + (Bx+C)/(quadratic)$	$4/x + (3-x)/(x^2 + x + 7)$
$(2x-3)$ in a bracket	kill it with $x = 1.5$, not 3	$4/(2x-3) - 1/(x-1)$
$(t-5)(t+1)$ with a story	ignore the story entirely	$A = 30, B = 4$
"is $(x-2)$ a factor"	sub $x = +2$. $f(a) = 0$ means yes	$f(2) = 0 \rightarrow$ yes
" $(x-3)$ is a factor, find a"	set $f(3) = 0$, solve	$9 + 3a - 15 = 0 \rightarrow a = 2$
Method: $\times$ by the whole bottom, then pick the x that kills a bracket. When that runs out, compare the number of x^2 or x on each side.		

ORDER OF OPERATIONS, AND EQUATIONS		
Brackets $\rightarrow$ pOwers $\rightarrow$ Divide and Multiply, left to right $\rightarrow$ Add and Subtract, left to right.		
$6 + 3 \times 10 = 6 + 30 = 36$ (not 9×10) $5 + 2^3 \div 4 - 1 = 5 + 8 \div 4 - 1 = 5 + 2 - 1 = 6$		
The power outranks the divide, and the divide outranks the add. Doing $5 + 8$ first gives $13 \div 4$, which is the trap.		
LOOKS LIKE	MOVE	WORKED
$3x - 7 = 6$	undo in reverse order	$x = 13/3 = 4.33$
$4x + 2 = 5x + 10$	collect x on the side that keeps it +ve	$x = -8$
$7t - (3 - t) = 2$	the minus flips EVERY sign inside	$8t - 3 = 2 \rightarrow t = 5/8$

$(9+5k)/4 - 3 = 2k$

multiply EVERY term by 4, both sides

$k = -1$

$5x + 2y = 13,$
 $x + 2y = 9$

a term matches $\rightarrow$ subtract

$x = 1, y = 4$

$2a + 5b = 10,$
 $3a - 4b = 15$

nothing matches $\rightarrow$ scale both, then add

$a = 5, b = 0$

"use substitution"

make the lone letter the subject, sub in

$x = -0.5, y = -6.5$

$-5c - 7 > 8$

$\div$ by a NEGATIVE turns the sign round

$c < -3$

$34 > p - 12$

flip the whole statement to put p on the left

$46 > p \rightarrow p < 46$

"between \$199 and \$299, \$26 off"

double inequality, same op on all three

173 to 273

Number line: **hollow** = not included, **solid** = included.

BEFORE YOU HAND IT IN

☐ Calculator in **DEG?** (radians only where there is a π)

☐ Every trig equation — did I give **both** answers?

☐ tan second answer is **+180**, not 360 minus

☐ Minus in front of a bracket — did every sign inside flip?

☐ Completed square with a number in front — did it multiply what I subtracted?

☐ Turning point — did I sub back into the **original**?

☐ 2nd derivative: +ve is a MINIMUM

☐ Did it want x, or the height / area / time?

☐ Indefinite integral — **+ c**? Definite — no c, upper minus lower?

☐ Area came out negative $\rightarrow$ I subtracted backwards, or it dips below the axis

☐ Line of symmetry written as **x = ...**, not a bare number

☐ 2 d.p. unless told otherwise. Units where asked.

☐ Anything blank? Write the first line anyway. It earns.

FIRST ORDER DIFFERENTIAL EQUATIONS — MATCHING ONLY

THE STORY	EQUATION	SOLUTION AND GRAPH
decays, falls towards nothing	$dx/dt = -kx$	Ae^{-kt} . drops, flattens onto 0
grows without limit	$dx/dt = kx$	Ae^{kt} . rises ever more steeply
cools towards a room at c	$dx/dt = -k(x-c)$	$c + Ae^{-kt}$. levels onto the line $x=c$

The give-away: rate depends on the amount *itself* $\rightarrow kx$. On a **difference** from a fixed level $\rightarrow -k(x-c)$.

Constants, in this order: **k** from the equation using the given rate; then **A** from $t = 0$ in the solution. $e^0 = 1$.

coffee 90° , room 20° , $k=0.01 \rightarrow T = 20 + 70e^{-0.01t}$

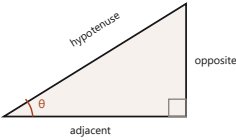
MCQ tie-break: a $-k$ in the equation **MUST** give a $-kt$ in the power. The trap option pairs a right equation with a mismatched solution.

"Time to reach room temperature" never happens — it is asymptotic. Say "approaches 20° "; from the graph about 450 to 500 min".

Ends at

nothing, nowhere, or flat on a line. That one question picks the row.

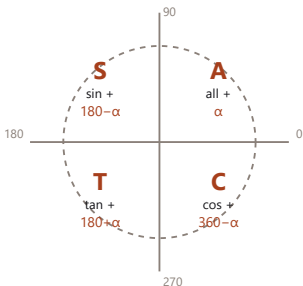
TURNING POINTS, AND WORD PROBLEMS		
1 differentiate	2 set = 0, solve for x	3 sub x into the ORIGINAL for y
2nd derivative +ve $\rightarrow$ MINIMUM. -ve $\rightarrow$ MAXIMUM. (+ve is a cup, it holds water, so it is the bottom)		
$y = 5 + 6x - 3x^2 \rightarrow 6 - 6x = 0 \rightarrow x = 1 \rightarrow y = 8 \rightarrow y'' = -6 \rightarrow$ MAX at (1, 8)		
$y = 7x^2 + 14x - 12 \rightarrow x = -1 \rightarrow y = -19 \rightarrow y'' = 14 \rightarrow$ MIN at (-1, -19)		
Free check: for any quadratic the sheet gives the turning point as $(-b/2a, c - b^2/4a)$. Same answer, five seconds, no calculus.		
Read what is wanted. Time, or height? x, or the area? Mortar: work out $100 \sin 30 = 50$ before differentiating $\rightarrow t = 5$ s. Fence: $x = 5$ but the answer is the area, 50. Blood alcohol at $t = 3$ is -0.02 — the minus IS the answer.		

TRIG — WHICH RATIO OR RULE?		
YOU ARE GIVEN	USE	WORKED
right angle, hyp given, want the side OPPOSITE	SOH: opp = hyp × sin	40°, hyp 8 → h = 8sin40 = 5.14
right angle, adj given, want the HYPOTENUSE	CAH rearranged: hyp = adj ÷ cos	30°, adj 15 → k = 15/cos30 = 17.32
right angle, angle + one leg, no hyp anywhere	TOA: opp = adj × tan	8 tan40 = 6.71
two sides, angle unknown	inverse: sin ⁻¹ (o/h), cos ⁻¹ (a/h), tan ⁻¹ (o/a)	
ladder, angle with the GROUND, foot 3 m out	ground = adjacent, ladder = hypotenuse	3/cos45 = 4.24 m
...“angle with the WALL”	no trig at all: 90 – that	→ 45°
pendulum L, angle from the VERTICAL	rise = L(1 – cosθ). check where the angle is marked	45(1–cos30) = 6.03 cm
NO right angle: 2 angles + a side, or an angle and its opposite side	SINE RULE (not on sheet) a/sinA = b/sinB	C=35, c=40, b=50 → B = 45.80°
NO right angle: 3 sides, want an angle	COSINE RULE (not on sheet) cosA = (b ² +c ² –a ²)/2bc	42, 37, 26 → A = 60.63°
NO right angle: 2 sides + the angle BETWEEN	COSINE RULE: c ² = a ² +b ² –2ab·cosC	60°, 8, 6 → c = 7.21
Which rule? An angle opposite a known side, or two angles → SINE. Three sides, or two sides with the angle wedged between → COSINE. In the cosine rule the side you square on the left is the one FACING the angle.		
 SOH sin = O/H CAH cos = A/H TOA tan = O/A opposite is ACROSS from θ, hypotenuse is always the long sloping		

TRIG EQUATIONS — ALWAYS TWO ANSWERS		
YOU HAVE	QUADRANTS	THE TWO ANSWERS
sin = +	1st, 2nd	α and 180 – α
sin = –	3rd, 4th	180 + α and 360 – α
cos = +	1st, 4th	α and 360 – α
cos = –	2nd, 3rd	180 – α and 180 + α
tan = +	1st, 3rd	α and α + 180 – NOT 360–α
tan = –	2nd, 4th	180 – α and 360 – α

α = inverse of the value with its **minus sign removed**. Always 0 to 90.
4sinθ+5=3 → sin=–0.5, α=30 → **210, 330**
4cosθ+2=5 → cos=0.75, α=41.41 → **41.41, 318.59**
2tanθ=27 → tan=4.5, α=77.47 → **77.47, 257.47**

If the bracket holds 2θ: solve for 2θ, get the two answers, **add 360 to each**, then divide by 2. Two cycles means **four** answers. Both sin and cos in one equation → collapse to one function with an identity (they are on your sheet) first.



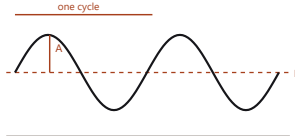
WAVES — READING AND BUILDING

Off a drawn graph, in this order: D the middle line → B count the cycles in 360 → A half the gap → then its **sign**.

At θ = 0 the curve is: on the midline going **up** → +sin · on the midline going **down** → –sin · at the **max** → +cos · at the **min** → –cos

max 4 min –2, 2 cycles → **y = 3sin(2θ) + 1**
max –1 min –3, 2 cycles, starts mid rising → **y = sin(2θ) – 2**
max 6 min –2, 3 cycles, starts mid falling → **y = –4sin(3θ) + 2**
max 2 min –8, 4 cycles, starts at the min → **K=–5, L=4, M=–3**

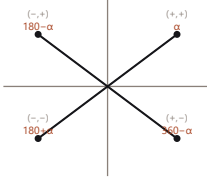
“0.5 cycles in 180” means B = 1, because B counts cycles per 360. **A right shift of 10 makes C = –10**, not +10.
Max and min values: max = |A| + D, min = –|A| + D. **B and C are irrelevant.** 3sinθ–4 → –1 and –7 · –2cos(3x)+5 → 7 and 3



A half the gap D the middle line B cycles in 360 C sideways
max = |A| + D min = –|A| + D. B and C never affect them.

VECTORS		
LOOKS LIKE	MOVE	WORKED
“magnitude”	√(x²+y²) — not on sheet. signs vanish	(–7, –11) → √170 = 13.04
“direction”	α = tan ⁻¹ (y / x), then place it ↓	see the quadrant box
p + q, p – q	row by row. same dimension only	(4, 12)+(2, –4) = (6, 8)
r – 3s	multiply FIRST, then subtract	→ (1, –9, –13)
(5, y)+(7, 2) = 3(2x, 2)	expand the scalar first → (6x, 6)	x = 2, y = 4
“column vector AB”	B – A. head minus tail	A(2, 7, 2) B(8, 4, 1) → (6, –3, –1)
PQ and QP	PQ = OQ – OP. QP is every sign flipped	(–5, 0) and (5, 0)
BX + XC	middle letters MATCH → add	→ (–1, 11), BC = 11.05
PX and CX	both end at X → PX – CX	→ (5, 1), PC = 5.10
“angle between”	dot, both magnitudes, then cos ⁻¹ . formula on sheet	(12, 5)(3, 7) → 44.18°
...dot comes out negative	that means the angle is OBTUSE. keep the minus	→ 139.40°
“are they perpendicular?”	dot = 0. no cos ⁻¹ needed	
a × b	formula on sheet. middle component reversed. check: dot with either gives 0	(8, 1, 2)×(2, 1, 5) → (3, –36, 6)

VECTOR DIRECTION — PLACE THE ANGLE



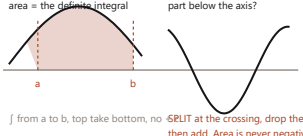
α = tan⁻¹(|y|/|x|), always 0 to 90. Sketch the arrow, then place it.

MATRICES		
LOOKS LIKE	MOVE	WORKED
“state the order”	rows × columns. two numbers, never multiplied out	[1 4 –2] → 1×3
“order of the product, if applicable”	INNER numbers must match. OUTER numbers give the answer	(3×1)(1×4) → 3×4
...when they do not match	say not possible — that IS the answer	(2×2)(1×3) → undefined
A + B, A – B	same order only. position by position	→ [[1, –2], [4, –3]]
kA	every entry, including zeros and negatives	0.5[[3, –4], [0, 2]] → [[1.5, –2], [0, 1]]
row × matrix	row of the left against column of the right, one at a time	[1 –2 5][...] → [5 –3 23]

2×2 × 2×2	layout on sheet. AB ≠ BA	→ [[29, –3], [38, –1]]
“= (m, 1), find m”	only compute the row you need; the given entry checks it	4(8)+2(9) → m = 50
“express in matrix form”	line the variables up in the same order, write 0 for any missing one	
determinant	ad – bc. formula on sheet. watch double negatives	$\frac{(-2)(-4)-(-2)(3)}{2} =$
inverse	swap a and d, negate b and c , ÷ det. det=0 → none	
“solve using the inverse”	U = M⁻¹V. the inverse goes on the LEFT	→ x = –39, y = 19

DIFFERENTIATE — READ THE FUNCTION		
LOOKS LIKE	MOVE	WORKED
ax ⁿ	power to the front, then n – 1	4x ⁵ → 20x ⁴
7x	the x vanishes	→ 7
a bare number	goes to nothing	98 → 0
6/x, 5/(12x ²)	REWRITE as a negative power	6x ⁻¹ → –6x ⁻²
√x, √[3]{x}	REWRITE as a fraction power	x ^{0.5} → 0.5x ^{-0.5}
() () two brackets	EXPAND. there is no product rule	(3x ² +5)(x ² +1) → 12x ³ +16x
(sum)/(one term)	SPLIT term by term, then rewrite each	(x ² +2x+5)/x → split first
A sin x, sin Bx, cos Bx, ae ^{kx+b}	on the sheet. copy it	12cos(x–3) → –12sin(x–3)
9 – cos5x	cos → –sin, and minus a minus is plus	→ +5sin5x
A sin(Bx+C)	not on sheet: AB cos(Bx+C)	3sin(2x–7) → 12x ³ +16x
ln(anything)	not on sheet: inside' OVER inside	ln(x ² +4x) → (4x ² +4)/(x ² +4x)
() ⁿ a bracket to a power	CHAIN: n()ⁿ⁻¹ × inside'	(x ² –4) ⁵ → 10x(x ² –4) ⁴
√(bracket)	rewrite as () ^{0.5} , then chain	→ 1.5x ² /√(x ² +2)
d ² y/dx ²	differentiate your own answer again	

INTEGRATE — READ THE INTEGRAND		
LOOKS LIKE	MOVE	WORKED
ax ⁿ	add 1 to the power, divide by the NEW power	–5x ² → –5x ³ /3 + c
a bare number	times x	50 → 50x + c
1/x ³ , 1/x ⁿ	REWRITE as x⁻ⁿ	→ –1/(2x ²) + c
1/x	the one the power rule fails on	→ ln x + c
a/(bx+d)	NOT on sheet: (a÷b)·ln(bx+d)	7/(3–4x) → –1.75ln(3–4x)+c
√x	rewrite x ^{0.5}	→ (2/3)x ^{1.5} + c
a sin kx, a cos kx	on the sheet. DIVIDE by k	sin9x → –cos9x/9 + c
a sin(Bx+C)	not on sheet: –(a/B)cos(Bx+C)	4sin(5x+7) → –0.8cos(5x+7)+c
sin x · cos x	REWRITE with sin2x = 2sinxcosx	4sinxcosx → –cos2x + c
ae ^{kx+b}	on the sheet. divide by k	e ^x → e ^x /4 + c
a sum of terms	one at a time, ONE +c at the end	
dy/dx AND a point given	integrate, sub the point, find c, write the WHOLE equation	c = –8.25
numbers on the f	no +c. upper minus lower	f ₁ ·3x = 54–1.5 = 52.5
“area bounded by...”	it is the definite integral	5x ² –2x+8, 1 to 3 → 108
curve dips BELOW the axis	split, drop the minus, then add	x ² from –3 to 2 → 24.25



f from a to b, top take bottom, –SPLIT at the crossing, drop the mir then add. Area is never negative.